

## Heat & Thermodynamics

**Heat and Temperature:** Heat and Temperature are two important terms in physics. If a piece of hot metal is dropped in to water in a glass, the temperature of water increases while that of metal decreases. This transfer of something from hot metal to cold water is a form of energy and this called Heat. We can say that, Heat is a form of energy whose absorption makes a body hot and rejection makes a body cold.

The feeling of hotness and coldness is called the Temperature. It is the thermal state of a body which controls the flow of heat from one body to another body and determines the direction of heat flow.

**Principle of temperature Measurements:** Any physical property of a substance which varies regularly with temperature and the variation is easily observable can be used to measure temperature. For example: increase in volume of a gas on heating, increase in pressure on heating, expansion of a liquid on heating, increase in resistance on heating etc are some of the physical properties which vary uniformly. These properties are employed to measure temperature.

The suitable physical properties employed to measure temperature are called **thermometric properties** and the substances whose thermometric properties are used to construct thermometers are called **thermometric substances**.

**Platinum Resistance Thermometer:** Generally, the resistance of a conductor increases with the increase of temperature. So, by knowing the resistance, the temperature can be found out. Thermometer constructed on this principle is called the resistance thermometer. Thermometer of this type made of platinum metal is called the Platinum Resistance Thermometer.

In 1871, Sir William Siemens first constructed the platinum resistance thermometer. Even today platinum is extensively used as resistance thermometer. Platinum is a very rigid metal and dependable. Its characteristics are not changed by repeated use.

The relation between temperature and resistance is given by,

$$R_{\theta} = R_0(1 + \alpha\theta)$$

Here,  $R_{\theta}$  = resistance of the platinum wire at temperature  $\theta$

$R_0$  = resistance of the wire at  $0^{\circ}$  temperature

$\alpha$  = a constant

Now can be written as,  $\theta = \frac{R_{\theta}-R_0}{\alpha R_0}$  ----- (1)

Let  $R_{100}$  is the resistance of the platinum wire at temperature  $100^{\circ}\text{C}$ . Now we get,

$$R_{100} = R_0(1 + \alpha \cdot 100)$$

$$\text{or, } \alpha = \frac{R_{100}-R_0}{100 \cdot R_0}$$

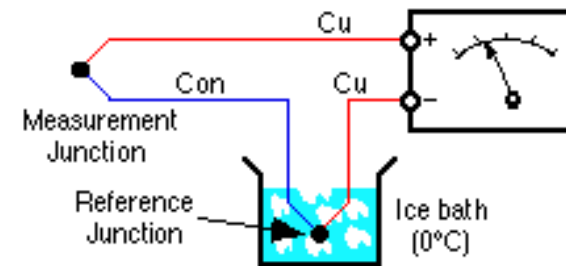
Using Eq.1; we get,  $\theta = \frac{R_{\theta}-R_0}{R_{100}-R_0} \times 100$

Platinum resistance can be used to measure temperature from  $-250^{\circ}\text{C}$  to  $1300^{\circ}\text{C}$ .

**Problem-1:** The resistances of a resistance thermometer at  $0^{\circ}\text{C}$  and  $100^{\circ}\text{C}$  are  $8\Omega$  and  $20\Omega$  respectively. If the thermometer is placed in furnace it shows  $32\Omega$  resistance. Find the temperature of furnace.

**Problem-2:** A platinum resistance thermometer reads  $2\Omega$  and  $2.73\Omega$  at ice point and steam point respectively. Find the temperature at which it reads  $4.83\Omega$ .

**Thermo-electric Thermometer:** It is based on the thermo-electric power. Two dissimilar suitable metallic wires (say Copper and Constantan) are joined together successively which form two junctions of a *thermocouple*. When these two junctions are placed at two different temperatures, a thermo *emf* is generated in the thermocouple, (Fig.1).



Traditional Thermocouple Measurement

By measuring the thermo *emf*, the difference of the temperatures between the junctions can be found out. Now, if the temperature of a junction is known, then the unknown temperature can be determined by the following equation-

$$e = a\theta + b\theta^2$$

Here,  $e = \text{thermo} - \text{emf generated in the thermocouple}$

$\theta = \text{temperature}$

$a$  and  $b$  are two constants to be determined from two fixed temperature.

**Pyrometer:** Thermometers which are used for measuring temperatures above  $500^{\circ}\text{C}$  are referred to as pyrometers and branch of physics in which temperatures are measured above  $500^{\circ}\text{C}$  is called Pyrometer Thermometry. According to the above definition, Platinum resistance thermometer, gas thermometer and thermo-couple thermometer are also called Pyrometers. They are respectively called resistance pyrometer, gas pyrometer and thermo-couple pyrometer. There are two types of pyrometers which measure temperature by the method of radiation.

They are-

- Total radiation pyrometer:
  - Thermometric Substance- Heat Radiation Energy
  - Thermometric Properties- Amount of Heat
  - Range of Temperature-  $500^{\circ}\text{C}$  and above
- Optical pyrometer:
  - Thermometric Substance- Light Energy
  - Thermometric Properties- Brightness
  - Range of Temperature-  $600^{\circ}\text{C}$  to  $1500^{\circ}\text{C}$